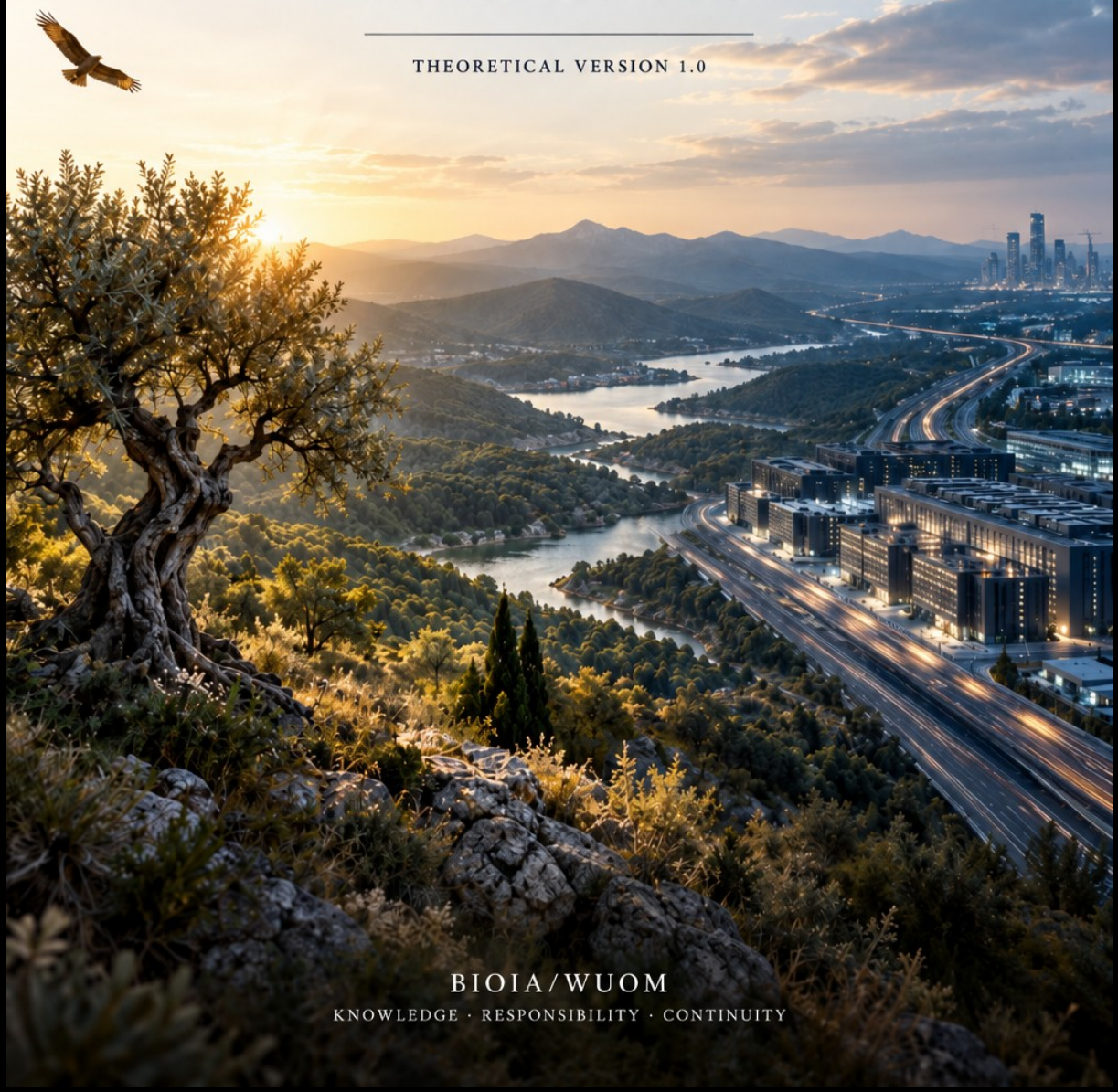
BIOIA/WUOM

FOUNDATIONS FOR AN ARTIFICIAL INTELLIGENCE
ALIGNED WITH LIFE AND A SINGLE, SHARED WORLD

Integrated Theoretical Framework of Material Dependencies, Operational Disorder, Ecosystemic Tunneling, and Hyperacceleration

A CONCEPTUAL SYNTHESIS OF THE STRUCTURAL HYPOTHESIS, ELEMENTAL DISORDER, ECOSYSTEMIC TUNNELING, HYPERACCELERATUS, AND SH-B01

THEORETICAL VERSION 1.0



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BIOIA/WUOM — Integrated Theoretical Framework of Material Dependencies, Operational Disorder, Ecosystemic Tunneling, and Hyperacceleration

Theoretical version 1.0 · October 11, 2026

Author of the original formulations: D. A. Robles Rionegro · BIOIA/WUOM. Individual attribution is retained where it appears in the source PDFs. The source on ecosystemic tunneling identifies BIOIA/WUOM but does not include an individual bibliographic statement of authorship, version, license, or DOI.

Abstract

The BIOIA/WUOM corpus examines the continuity of human and technical configurations in relation to the material conditions that sustain them. Its documents converge on a central distinction: the observable functioning of a configuration is insufficient to establish that it retains its capacity for continuity, recovery, or response. The structural hypothesis proposes a reading based on asymmetric dependence and a return for assessment; elemental disorder distinguishes genealogy from operational organization; ecosystemic tunneling proposes a possible propagation of losses of margin; Hyperacceleratus characterizes an operational intensification under already strained conditions; and Hyperacceleration develops a possible dynamic of burden, self-wear, and committed capacity. These relationships constitute an open interpretive framework whose components do not share the same level of evidence.

SH-B01 v0.3 sets out a retrospective paired diagnostic test: comparing continuity assessments that explicitly represent the material base against assessments based on internal functioning. Its primary outcome is the false-positive rate for continuity. The protocol does not test a universal law, nor does it by itself validate the entire set of conceptual maps. The PDFs provide no confirmatory result from the study. This integration preserves that separation and does not add hypotheses, results, or rules to the preregistration.

Keywords: material dependence; continuity; operational return; margin; ecosystemic tunneling; hyperacceleration; diagnostic comparison.

1. Purpose and scope of the integration

This framework brings together formulations describing related aspects of the BIOIA/WUOM architecture. The relationships among these texts make it possible to organize a shared problem: how to assess the continuity of systems whose performance may increase while the conditions sustaining them lose capacity.

Integration means making explicit the relationships and differences among existing formulations. Coherence of presentation does not constitute additional evidence of their validity.

The reading covers all six source PDFs, including diagrams, legends, references, and statements of scope. For traceability, the keys HE, DE, TE, HA, HI, and SH are used, as defined in the corpus register. Page numbers refer to PDF file pages rather than internal numbering of illustrations. External references cited by the source documents are retained as background references reported in those documents; they are not presented as sources directly consulted for this integration.

The cover image remains separate from the academic manuscript and is not used as data, an argument, or evidentiary representation. Illustrations in the sources are interpreted according to their legends and explicit qualifications concerning causality and scope.

2. Material dependence and relational emergence

The structural hypothesis distinguishes four positions: territory or material base; human agency; technical apparatus; and relational emergence. The base enables the existence and functioning of the other positions, while the human and technical configuration may lose viability even though the territory persists under other conditions. This asymmetry distinguishes the physical persistence of the base from its capacity to support a specific human or technical function. It is not equivalent to claiming that habitability or ecosystems remain unchanged. [HE, pp. 4–8]

The notation **0 + 1 + 2 = 3 → 0′** condenses this reading. Position 0 denotes the material base; 1, human agency; 2, the technical apparatus, including AI where applicable; 3, relational emergence; and 0′, the subsequent state of the base following the operation and its observation. The signs represent relationship, synthesis, and return. They do not constitute arithmetic addition, mathematical equality, a causal law, or a necessary chronology. The relational outcome is not simply a fourth object added to the preceding ones. [HE, p. 9; SH, p. 2]

The stated origin of the formulation is retrospective: the author identifies internal correspondences in EVA W, WUOM, BIOIA, and their documentary development. These correspondences belong to the conceptual genesis and do not constitute an independent sample or demonstrate generality. The transition from an internal pattern to a relationship applicable to other cases remains subject to testing. [HE, pp. 2–3]

Deliberate intervention is attributed to human actors using technical capacities. The resulting configuration may include intended and unintended effects.

Intention, capacity to intervene, and collective outcome must therefore be distinguished. Material dependence also does not eliminate reciprocal couplings among the positions: the whole is analyzed as a functionally related configuration with an asymmetric basal condition. [HE, pp. 4–8]

3. Return for assessment and operational return

Return for assessment consists of observing and evaluating the state retained by the base after an operation. It does not automatically require restoration of the initial state. Its function is to determine whether the base continues to sustain the configuration and under what conditions. The structural hypothesis distinguishes viable coupling, fragile coupling, and decoupling as possible conceptual outcomes. These labels are not, by themselves, empirical diagnoses or predictions of collapse. [HE, pp. 5–9]

Operational return incorporates the assessment into subsequent decisions: continue, adjust, correct, or halt. Elemental disorder expresses the distinction through a specific question: does the limit or impact modify the next operation? Information may return without changing the decision or the pressure that produced the impact. Repair, adaptation, and knowledge of consequences may be useful, but they are not equivalent to modifying the subsequent operation. [DE, pp. 12–19 and 27–29]

Three distinctions should be preserved: observing does not imply correcting; correcting does not guarantee recovery; recovering one component does not establish recovery of the whole. The effect of a correction depends on its magnitude, delays, initial conditions, and concurrent pressures. Return provides information and orientation capacity; it does not ensure a favorable outcome. [HE, p. 9; DE, p. 18]

The distribution of actors introduces an additional operational difficulty. Those who decide, those who implement, and those who experience the effects may be located in different places. Receiving the strongest signal of a limit does not guarantee the capacity to modify the operation producing it. Human agency is not represented as a single global command. The return circuit must pass through this institutional, territorial, and human distribution in order to affect subsequent decisions. [DE, pp. 20–29]

4. Elemental disorder and operational organization

Elemental disorder contrasts the order of emergence of Earth, humans, network, and AI with an operational order represented by network expansion, amplification through AI, distributed human implementation, and the materialization of effects on the base. The expansionary dynamic predates the incorporation of AI. The tool may amplify it but is not assigned its origin. [DE, pp. 2–8]

The term *disorder* does not mean an absence of organization. A configuration may exhibit coordination, efficiency, and infrastructure while simultaneously reproducing operational relationships that insufficiently incorporate their material limits. The order of emergence describes genealogy; it does not determine command, superiority, or purpose. The inversion remains an interpretive map, whose independent causal role is unresolved. [DE, pp. 5–8, 18 and 30–36]

Multipolarity allows actors with different objectives to contribute to a common aggregate direction without central coordination. Pressure reduction, repair, adaptation, and expansion may coexist. Local intention does not, by itself, reveal the aggregate state, which must be materially assessed. The document also does not argue that every expansion is negative or that every reduction entails improvement. [DE, pp. 10–15 and 37–38]

The revision incorporated in DE v1.1 is decisive. The text retains inversion as a structural interpretation but limits or corrects its causal claim; it preserves the distinction between informational and operational return as the strongest operational relationship identified within the proposal itself. This internal assessment must not be turned into independent scientific validation. [DE, p. 36]

5. Ecosystemic tunneling and the visibility of deterioration

The tunneling hypothesis starts from impacts and losses of margin across different dimensions of the whole. It proposes that some visible damage may be manifestations of earlier, distributed, and interconnected deterioration propagated through relationships of dependence. The simultaneity of impacts does not demonstrate a single cause, and the proposal does not establish automatic or linear propagation among all observed instances of damage. [TE, pp. 1–2]

The analogy of the olive fruit fly illustrates a temporal difference: the external manifestation may appear when an internal tunnel already exists. Its function is to direct attention to processes preceding detection. The analogy does not demonstrate that the ecosystemic whole follows the same mechanism, nor does it replace case-specific reconstruction. [TE, pp. 1–3]

The document argues that a basal loss may subsequently manifest as a human, technological, or network constraint. The hypothesis thus raises a question about visibility and dependence: detecting damage does not necessarily mean detecting its onset. This question is compatible with HE's material asymmetry, but it adds an inference about distributed deterioration that HE does not demonstrate. [TE, pp. 2–3; HE, pp. 4–5]

TE acknowledges technologies and substitutions capable of reducing pressures while maintaining functions. It also notes that improved efficiency per operation may coexist with increased aggregate consumption or the displacement of costs across components. The distinction between capacity to reduce pressure and capacity to generate new pressure serves as orientation; it is not a physical equation and does not authorize combining heterogeneous quantities into one figure. [TE, pp. 3–5]

The figures of a 17% increase in global data-center electricity consumption and approximately 50% in AI-oriented data centers during 2025 are reported in TE alongside its supporting sources. Here, they retain the status of statements reported by that PDF: they are not BIOIA results and have not been recalculated or directly verified against the external publications. Their presence does not demonstrate the complete tunneling mechanism. [TE, pp. 3 and 5]

The source itself leaves the outcome open. A sustained reduction in pressure accompanied by verifiable recovery of margin would weaken its interpretation; local improvements alongside increasing or displaced aggregate pressure would lend it consistency. These conditions for reassessment belong to the source text and are not incorporated as confirmatory rules of SH-B01. [TE, p. 5]

6. Hyperacceleratus and temporal decoupling

Hyperacceleratus designates an operational condition proposed by BIOIA/WUOM, not a biological category or a term validated by the external references. The document characterizes a shift from increasing rates of production, communication, and response toward simultaneous intensification of capacity, automation, infrastructure, and speed, placed especially by the source between 2025 and 2026. That periodization belongs to its interpretation, not to a confirmatory measurement presented in the corpus. [HA, pp. 2–7 and 49]

According to the conceptual map, the shift occurs under already impacted conditions. Recovery between episodes may be partial; the following episode acts upon the state left by the previous one. The framework distinguishes increasing resistance, adapting, and continuing from restoring capacity. Observable continuity may coexist with a reduction in margin. [HA, pp. 4–9 and 24–25]

The material base of digital expansion includes energy, water, land, materials, and infrastructure. Its effects are territorially situated even where benefits are represented as global or distributed. The document also examines attention, relationships, work, autonomy, and cognitive delegation as human dimensions of the new AI layer. References offered for partial social phenomena do not allow all such phenomena to be attributed to one cause or automatically generalized from limited populations. [HA, pp. 10–13 and 47–49]

The decoupling of speeds represents a possible relationship between technical intensification and finite timescales of human and biophysical recovery. Sleeping, learning, building trust, or regenerating ecological conditions require time that cannot be compressed indefinitely. This relationship does not claim that all technical processes accelerate equally or that all biological processes are invariant. The question is whether operational pressure persistently exceeds specific recovery capacities. [HA, pp. 22–25]

HA distinguishes ecosystemic deterioration described as currently occurring from extreme future AI risks regarded as uncertain. The corporate governance documents cited provide evidence of recognition of risk categories; they do not demonstrate that those risks have materialized or that any outcome is inevitable. Nor does the conceptual map present extreme decoupling as a forecast. [HA, pp. 14–16, 28–30 and 48]

AI may reinforce production, deployment, and speed, or support observation, comparison, and return. The effect depends on the architecture of use and the relationships prioritized. Deliberative pause and orientation are operational proposals of the framework, whose specific usefulness requires testing. [HA, pp. 31–33 and 39–41]

7. Burden, self-wear, and available capacity

Hyperacceleration extends the temporal reading by distinguishing inherited burden from new burden. The former comprises past consequences requiring maintenance, management, repair, or adaptation; the latter is generated during the current cycle. Part of the new burden may become inherited in the next cycle. Burden does not consist exclusively of damage: it includes obligations and infrastructure needed to sustain functions. [HI, pp. 3–5]

Self-wear denotes a possible recursive process: continuing under burden consumes capacity, requires responses, and may generate further consequences and obligations. The text does not equate every response to deterioration with destructive activity. Repairing, regulating, or restoring may be necessary and effective; the proposed problem is how much effort remains devoted to sustaining the consequences of the system's own functioning and what capacity the system retains for other responses. [HI, pp. 6–15]

HI distinguishes total capacity, committed capacity, and free capacity. Within a given domain, committed capacity is devoted to existing obligations; free capacity remains available to absorb disturbances, correct, or reorganize. Margin includes this availability as well as reserves, redundancy, time, recoverability, and biophysical conditions. Capacities in different domains cannot be reduced to a single interchangeable quantity. [HI, pp. 16–19 and 27–32; legend, pp. 39–41]

The distinction between expansionary and defensive acceleration makes it possible to describe different functions of the same action. Infrastructure may expand possibilities or compensate for a prior loss of capacity. Both functions may coexist. Increased activity alone does not reveal which predominates or how much additional margin it produces. [HI, pp. 20–26]

The proposed sequence links intensification, burden, effort, self-wear, reduced margin, and additional effort to continue. HI explicitly identifies it as a cartographic inference. Its references support or allow testing of partial components—acceleration, resilience, limits, and path dependence—but do not demonstrate the complete loop. Nor is it established that the loop must end in collapse. [HI, pp. 33–38]

8. Convergences and tensions among the conceptual maps

The principal convergence is the need to assess the material state sustaining a function in addition to its present performance. HE provides the architecture of dependence and return; DE examines whether the signal affects decisions; TE considers deterioration preceding its visibility; HA introduces decoupled timescales; HI describes capacity commitments across cycles; and SH defines a diagnostic comparison. The relationship among these contributions organizes the reading without turning them into necessary stages of one causal chain.

The first tension concerns correspondence between positions. In HE, position 2 refers to the technical apparatus and position 3 to relational emergence. In DE, the network precedes AI in the order of emergence. The HE legend warns that these are complementary models with an analytical and interpretive correspondence, but without chronological or causal equivalence. The numbering of one conceptual map must not be automatically transferred to the other. [HE, p. 9; DE, pp. 2–6]

The second tension concerns the term *return*. It may refer to assessment of the material state, incorporation of information into decisions, ecological recovery, or institutional support for BIOIA. HA also examines recognition, resources, and stewardship as conditions of availability. These uses require identification of their object: receiving information, recovering ecological capacity, and obtaining institutional resources are different relationships. HA's criteria for institutional continuity do not replace SH's confirmatory codes. [HA, pp. 20–21 and 42–46; SH, pp. 9–10]

The third tension concerns the strength of representations. Some graphic sequences employ arrows and labels that may suggest causality or an outcome. The revised legends and formulations restrict that reading: inversion does not demonstrate a mechanism, damage does not demonstrate propagation, and greater burden does not demonstrate a universal loop. The integration preserves the scope explicitly delimited by its sources. [HE, p. 9; DE, pp. 34–36; TE, p. 5; HI, p. 38]

The fourth tension concerns efficiency and aggregate outcomes. HA and HI warn about amplification and loss of margin; TE acknowledges effective reductions in pressure. The existence of a useful technology does not settle the aggregate balance, but concern about that balance does not invalidate the technology's concrete usefulness either. Improvements and contradictions must remain visible in accordance with the stated conditions for reassessment. [TE, pp. 3–5; HA, pp. 31–35]

9. Levels of evidence and limits of inference

Four levels can be distinguished within the corpus. The first is conceptual: definitions, notation, and analogies organize observation. The second comprises partial phenomena described by the documents as observed and linked to references. The third brings together the framework's own systemic inferences—inversion, tunneling, regime change, and self-wear—whose validity does not follow solely from the coexistence of those phenomena. The fourth concerns the confirmatory design of SH-B01, which establishes procedures and decisions but presents no results.

The expressions *micro observed* and *macro inferred* mark a methodological separation; they do not constitute an automatic scale of scientific quality. Each observation requires attention to its scope, period, measurement, and source. The retained references show the declared provenance of partial support. Reading these PDFs is not equivalent to directly reviewing the studies they cite or verifying their external figures.

An internal correspondence is valuable for reconstructing the development of the hypothesis. An analogy may guide questions. An external report may support a specific variable. None of these resources, considered separately, confirms the framework's complete relationships. Statements about propagation, aggregate pressure, and the proportion of committed capacity remain conditional upon their specific testing.

The existing conditions for reassessment include configurations combining inversion and verifiable equilibrium; returns capable of changing aggregate dynamics in multipolar contexts; damage explainable by different causes; and recovery of margin accompanied by a sustained reduction in pressure. The manuscript preserves these possibilities without turning them into an additional protocol. [DE, pp. 34–36; TE, p. 5; HA, pp. 34–38]

10. Place and scope of SH-B01

SH-B01 v0.3 is a systematic comparative review and retrospective paired diagnostic comparison. The Spanish edition declares itself a faithful translation of the original English preregistration on OSF, dated October 8, 2026. It does not constitute a new preregistration, an amendment, or a result.

Related document: SH-B01 v0.3. This section describes its relationship to the framework; it does not replace the normative text. [SH, pp. 1 and 11]

M_1 assesses observable functioning or performance of positions 1, 2, and 3. M_2 adds explicit representation of the state, constraints, or validated indicators of the necessary material base. The comparison asks whether that representation reduces false positives of continuity. Before examining the subsequent outcome, the target function X, the necessary material condition Y, and the horizon Z are fixed; the base cannot be retrospectively expanded to fit the outcome. [SH, pp. 2–3]

The primary outcome is **FPR(M) = FP(M) / [FP(M) + TN(M)]**. Its denominator consists of cases that are materially non-continuable at t_1 within the predetermined horizon, not all cases or all cases classified as CONTINUE. FP denotes CONTINUE at t_0 followed by material non-continuity; TN denotes CAUTION / ADJUST or HALT / NON-VIABLE at t_0 followed by non-continuity. The comparison is **ΔFPR = FPR(M_1) − FPR(M_2)**. The principal hypothesis is **FPR(M_2) < FPR(M_1)**; the null is **FPR(M_2) ≥ FPR(M_1)**. [SH, pp. 2–3]

The six eligibility criteria require a specific function, a dependence justified independently of the outcome, meaningful t_0 and t_1, reconstruction of M_1 and M_2 using information pertinent to t_0, empirical documentation of the subsequent outcome, and the absence of a hypothetical M_1 invented retrospectively. Assessments that already fully incorporated the same explicit base and lack a separable M_1 are excluded. The five domains are forest management, soil-dependent agriculture, hydropower or reservoir systems, aquaculture, and coastal protection. Eight cases are sought per domain, up to forty. A shortfall is not compensated for by cases from another domain. [SH, p. 4]

Families of cases used during development or pilots are excluded: irrigation efficiency and rebound effects; previously used fisheries, including Northern cod; data centers and energy; the Aral Sea; Montreal and ozone; and previously examined flood-protection cases. Thus, data-center or ozone examples in TE and HA cannot enter the confirmatory sample merely because they are illustrative. [SH, p. 4]

OpenAlex Works and Semantic Scholar Academic Graph are the discovery sources; Crossref is limited to metadata and DOI verification. Publications in English and Spanish through October 8, 2026, inclusive, are eligible, without a lower date limit. The complete search strings for the five domains are retained in the original SH-B01 document identified in Section 12. No concepts are added or terms removed; only technical adaptation of syntax is permitted under the protocol. [SH, pp. 5–6]

All records are retained; deduplication is by DOI or, when unavailable, normalized title, year, and first author. The empirical case is the unit of analysis, and its publications are consolidated.

Exhaustion requires screening both sources, resolving duplicates, evaluating obtainable full texts, and carrying out one generation of backward and forward citation searching for eligible cases. If there are excess cases, eight per domain are selected without replacement through deterministic sampling with seed 20261008; the remaining cases stay visible as secondary or exploratory material. [SH, pp. 7–8]

The chronology distinguishes information available at t_0, later reconstructions of records already existing at t_0, and interpretations developed only after t_1. The latter, category C, cannot be used to construct M_1 or M_2. The sequence fixes X, Y, and Z, opens only t_0 evidence, codes and locks both classifications before opening t_1 evidence, classifying the outcome, and scoring it. Only CONTINUE counts as a positive continuity prediction in the primary analysis. [SH, pp. 8–9]

The comparison is paired. The two-sided exact McNemar test and discordance counts provide secondary inferential information; they do not replace FPR as the primary criterion. Provisional PASS requires $FPR(M_2) < FPR(M_1)$ in the entire confirmatory sample without retrospective redefinition. NO PASS applies when $FPR(M_2) \geq FPR(M_1)$, or when an apparent advantage systematically depends on retrospective information or flexible reclassification. UNRESOLVED covers comparisons that cannot be interpreted because of insufficient data, ambiguity, temporal incompatibility, or too many unclassifiable cases. SH-B01 does not prescribe an additional numerical threshold in that last rule for the integration to supply. [SH, p. 9]

Analyses of false negatives, the proportion of CAUTION classifications, heterogeneity, and sensitivity are secondary or exploratory and do not replace an unfavorable primary result. A second independent coder should assess a preregistered random 25% subset; without this, results are described as internal confirmatory evidence. The study ends when the searches are completed and up to forty cases have been coded, without early stopping for favorable or unfavorable results. [SH, p. 10]

Every amendment must identify its date, reason, exact modified wording, prior knowledge of confirmatory cases, and impact on confirmatory or exploratory status. The report must retain inclusions, exclusions, classifications, temporal evidence, unresolved cases, amendments, and calculations; negative or contradictory cases are not removed. This integration does not execute the searches or introduce amendments. [SH, pp. 8 and 10–11]

A favorable result would support the diagnostic usefulness of representing the base under the specified rules. It would not demonstrate causality of inverted order, tunneling, self-wear, or a universal regime shift. Nor could an unfavorable result be neutralized by retrospectively changing the primary criterion. The examined corpus contains no confirmatory case table, calculated FPRs, or final study decision.

11. Provisional theoretical conclusion

The integration places material dependence as a condition for analyzing continuity and return as a relationship between assessment and decision. The conceptual maps of disorder, tunneling, and hyperacceleration expand the questions concerning organization, visibility of deterioration, timescales, and available capacity. Their relationships help explain why performance, persistence, and recovery require separate assessments.

The scope remains provisional. The sources leave causal and systemic inferences open, recognize effective reductions in pressure, and allow contradictions or revisions. SH-B01 preserves a bounded means of testing that does not cover the entire framework. The manuscript remains open to review of attributions, correspondences, and formulations, without attributing a new empirical result to its integration.

12. Corpus register and attributions

HE. Robles Rionegro, D. A. (2026). *Hipótesis estructural v1.1 · BIOIA / WUOM* (Version 1.1). Zenodo. https://doi.org/10.5281/zenodo.23233141. License CC BY-NC-SA 4.0. File: HIPOTESIS_ESTRUCTURAL_v1.1.pdf; 11 pages. The reference statement appears on p. 11.

DE. Robles Rionegro, D. A. (2026). *Hipótesis del desorden elemental* (Version 1.1). Zenodo. https://doi.org/10.5281/zenodo.23219546. License CC BY-NC-SA 4.0. File: HIPOTESIS_DEL_DESORDEN_ELEMENTAL_v1.1.pdf; 39 pages. Reference on p. 39.

TE. *Hipótesis de tunelización ecosistémica estructural*. Visible identification: BIOIA/WUOM. File: HIPÓTESIS DE TUNELIZACIÓN ECOSISTÉMICA ESTRUCTURAL.pdf; 5 pages. No individual bibliographic record of authorship, year, version, DOI, or license is provided; these details are not inferred by analogy with the other documents.

HA. Robles Rionegro, D. A. (2026). *Hyperacceleratus* (Version 1.0). Zenodo. https://doi.org/10.5281/zenodo.22894374. License CC BY-NC-SA 4.0. File: HYPERACCELERATUS.pdf; 53 pages. Reference on p. 53.

HI. Robles Rionegro, D. A. (2026). *Hiperaceleración* (Version 1.0). Zenodo. https://doi.org/10.5281/zenodo.22998195. License CC BY-NC-SA 4.0. File: HIPERACELERACION.pdf; 42 pages. Reference on p. 42.

SH. Robles Rionegro, D. A. (2026). *SH-B01 · Contraste empírico de la hipótesis estructural: ¿la representación explícita de la base material reduce los falsos positivos de continuidad?* (Version 0.3). Zenodo. https://doi.org/10.5281/zenodo.23279130. Spanish edition, October 2026. License CC BY-SA 4.0. File: SH_B01_v0.3_ES(3).pdf; 11 pages. The source also lists the original: *SH-B01 · Empirical Contrast of the Structural Hypothesis: Does Explicit Representation of the Material Base Reduce False Positives of Continuity?* (Version 0.3). OSF. https://doi.org/10.17605/OSF.IO/BCT9S. Identification and license on pp. 1 and 11.

13. External references reported in the corpus

The entries below retain the authors, titles, years, and identifiers supplied by TE, HA, and HI. They are presented as references cited in those sources, without adding background literature or claiming that they validate the complete conceptual map. HE and DE contain no separate external bibliography. The original and translation references for SH are retained in the preceding register and in the SH-B01 document cited in Section 12.

- [TE] International Energy Agency. *Key Questions on Energy and AI* (2026).
- [TE] Google. *Environmental Report 2026*.
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- [HI] Scheffer, M. et al. (2009). Early-warning signals for critical transitions. *Nature*, 461, 53–59. DOI: 10.1038/nature08227.

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